Taxi Trip Board Report: September 2026 Sample

Seven questions answered from a sample of 10,000 trips (Sept 1 to 30, 2026). 9,269 trips remain after cleaning. Prepared for the board. Figures are from the sample, not the full city data.

Read first: four claims to handle carefully

Claim that could embarrass us	What the data says
Dana's flyer: "the average ride brings in $30"	Not supported. Average driver income per ride is $27.29 and the typical ride (median) is $16.61. Only 33% of rides reach $30. The $30 figure is close to the average total charge, which includes a $0.50 fee the data cannot explain and which is not driver income. Do not use it.
A fixed $50 price for O'Hare to downtown	Do not use. On 97% of O'Hare-to-downtown rides of 10+ miles, the meter charged more than $50. The price should be $59 (Section 5).
"Cash riders never tip"	We cannot test this. The system does not record cash tips. The only two recorded cash tips look like entry errors. Say "cash tips are not recorded," never "cash riders never tip."
Airport advantage (Section 3)	Cleaning changed this result. Use the cleaned figures: airport rides earn $19 per hour more than city rides, not the $13 the raw data suggested.

Summary of answers

#	Question	Short answer
1	Does the average ride bring in $30?	No. Median $16.61; average $27.29. About 1 in 3 rides reach $30.
2	Where do rides start, and when are we busiest?	Near North Side, O'Hare and the Loop account for 59% of trips. Weekday demand peaks 8 to 9 am and 2 to 6 pm.
3	Which kind of ride is better for a driver?	Airport rides, by about $19 per hour (28%). A queue wait of 11 to 44 minutes per airport ride would erase it.
4	Which payment methods, and what should the tip screen suggest?	Mobile 37%, card 36%, cash 19% of rides. Card tippers give a median 23%. Offer percentage buttons, not a flat $3.
5	Is a fixed O'Hare-to-downtown price safe?	Not at the average or at $50. Price at $59. The 75th percentile metered fare is $57.50.
6	What can we tell riders about travel time to the Loop?	Most trips take 33 to 59 minutes. Nine in ten finish within 73 minutes. Evenings are slowest.
7	What problems did we find, and what did we do?	20 problems logged. 731 trips removed. Four cannot be fixed from this data and are stated as limits (Section 7).

Definitions used throughout

- **Driver income per ride** = fare + tips. Tolls, extras and total charges are excluded, per the board decision.
- **Downtown** = Near North Side (community area 8), Loop (32) and Near South Side (33).
- **Airport ride** = a trip that starts at O'Hare (76) or Midway (56). **City ride** = a trip that neither starts nor ends at an airport.
- **"Better" for a driver** = income per hour on the meter (total income divided by total trip hours). It excludes waiting and empty driving.
- **Payment types** are as the system records them. Cash tips are not recorded.
- **Statistical terms:** the *median* is the middle ride (half earn more, half less). A *percentile* is the value below which that share of rides fall: the 25th percentile is the point 1 in 4 rides fall below, the 75th is the point 3 in 4 fall below. The *middle 50%* is the range between the 25th and 75th percentiles. A *90th percentile* is the value 9 in 10 rides fall below. A *95% confidence interval* is the range the true value most likely falls in, given the sample.

1. Does the average ride bring in $30?

Answer. No. The average driver earns $27.29 per ride, but the typical ride earns $16.61. Only 33% of rides reach $30. The $30 figure matches the average total charge, which includes a $0.50 fee that is not driver income. I recommend the flyer does not use it.

Definition. Driver income = fare + tips, for the 9,266 rides with a recorded fare.

Figure 1. Driver income per ride, September 2026 sample (fare + tips, USD). The orange dashed line is the average; the black line is the typical ride (median).

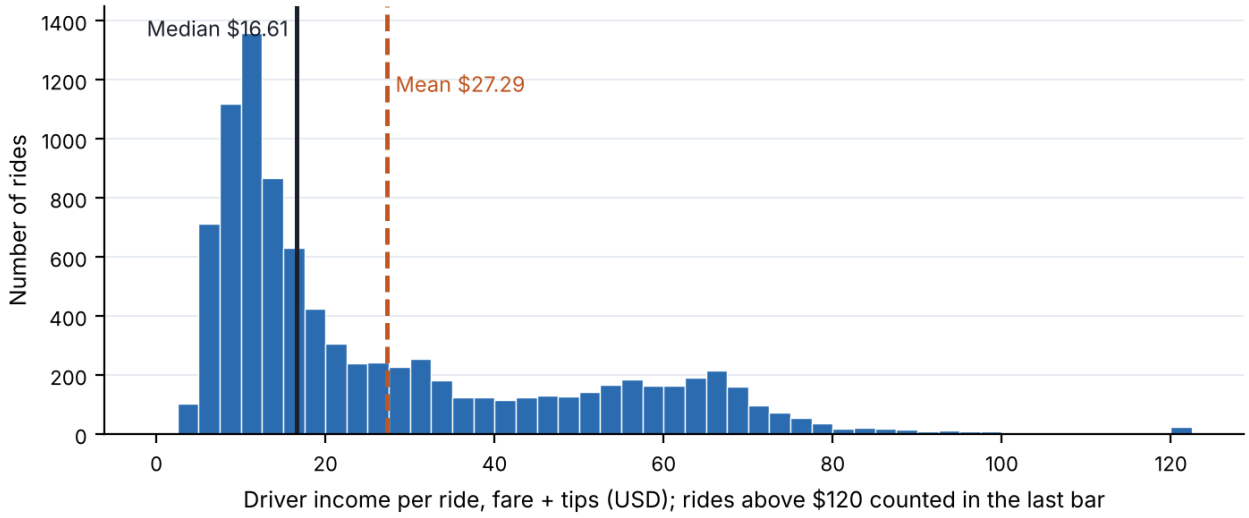


Table 1. Income per ride, September 2026 sample (USD per ride)

Measure	Value	What it means in plain words
Average (mean)	$27.29	Total income divided by number of rides. Pulled up by long, expensive rides.
Typical ride (median)	$16.61	Half of rides earn more than this, half earn less.
25th percentile	$10.64	One ride in four earns less than this.
75th percentile	$40.76	Three rides in four earn less than this.
Share of rides at $30 or more	32.6%	About 1 ride in 3.
Average without the top 1% of rides	$26.40	Removing the most expensive 1% only lowers the average by about $0.90.
Average total charge (raw data)	$31.19	Fare + tips + tolls + extras + the unexplained $0.50 fee. Not driver income. This is the likely source of "$30."

- **Why the average is higher than the typical ride:** the distribution has a long tail of expensive rides, not just a few outliers. The top 10% of rides account for 28% of total income.
- **Suggested flyer wording:** "The typical ride earns a driver about $17 in fares and tips. About 1 in 3 rides earns $30 or more."

What the data cannot tell us. Cash tips are not recorded, so income for cash rides (19% of trips) is understated. It excludes fuel, vehicle costs, taxes, and time between fares. It covers one month in a sample of unknown design. The board would need cash tip estimates and cost data before quoting driver earnings as a net figure.

2. Where do rides start, and when are we busiest?

Answer. Three areas account for 59% of trips: Near North Side, O'Hare and the Loop. On weekdays, demand peaks at 8 to 9 am and again from 2 to 6 pm, with 5 pm the busiest hour. On weekends, demand is steadier from 11 am to 6 pm. Mornings start most often in Near North, Near West and the Loop; evenings in Near North, O'Hare and the Loop.

Definitions. Pickup = the community area where the trip started. Shares are of all 9,269 cleaned trips. Blank pickups are labeled "Outside Chicago," which is an assumption (see below). Weekday = Monday to Friday. Morning = 6:00 to 10:59 am. Evening = 4:00 to 7:59 pm.

Table 2. Top 10 pickup areas (share of all trips, %)

Rank	Pickup area	Share of trips
1	Near North Side	23.0%
2	O'Hare	20.7%
3	Loop	15.5%
4	Near West Side	10.7%
5	Lake View	4.6%
6	Near South Side	3.4%
7	Lincoln Park	2.7%
8	Garfield Ridge	2.6%
9	Uptown	2.2%
10	West Town	1.7%

Figure 2. Share of trips by hour of day, weekdays vs weekends (% of that day type's trips).

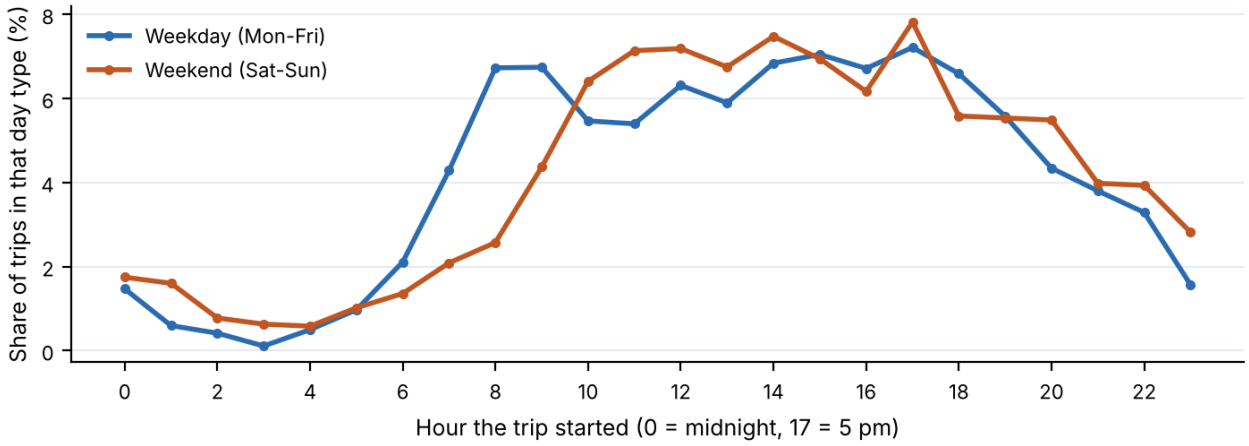


Table 3. Where to station drivers, by window (% of trips in the window)

Window	Top three pickup areas (share of trips in that window)
Morning, 6 to 11 am	Near North Side 21.5%; Near West Side 17.0%; Loop 15.8%
Evening, 4 to 8 pm	Near North Side 24.4%; O'Hare 22.1%; Loop 18.1%

What the data cannot tell us. Blank pickups (2.6% of raw trips) may be missing records rather than trips from outside Chicago. Trip counts show where riders were picked up, not where they wanted to go. We have no record of unmet demand, when drivers were idle, or where cabs were positioned. The board would need a log of ride requests and driver locations to plan station changes.

3. Which kind of ride is better for a driver, and by how much?

Answer. Airport rides are better. A driver earns about $87 per hour on the meter from an airport ride, against $68 from a city ride, about $19 more per hour (28%). Waiting in the queue erases that gap: 11 minutes of wait per airport ride if city drivers are always busy, 27 minutes if they are busy 75% of the time, and 44 minutes if they are busy 60% of the time.

Definitions. Airport ride = pickup at O'Hare (76) or Midway (56). City ride = neither end at an airport. 299 trips from the city to an airport are in neither group and are left out. "Better" = income per hour on the meter. "Busy" = share of paid working time spent with a fare on board; the rest is empty driving.

Table 4. Airport vs city rides (USD unless stated)

Measure (cleaned data)	Airport pickup	City ride	Difference
Rides	2,157	6,810	
Income per ride, average (USD)	$56.26	$16.79	$39.46
Trip length, average (minutes)	38.7	14.8	+23.9
Income per hour on the meter (USD/hour)	$87.29	$68.23	$19.07

Figure 3. Driver income per paid hour: airport pickups vs city rides at different busy shares (USD per hour).

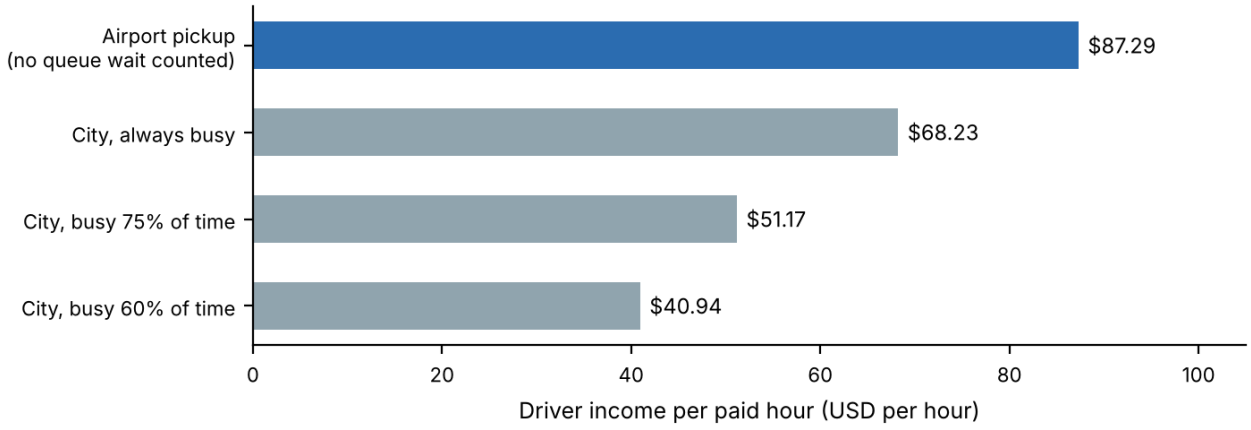


Table 5. Queue wait that erases the airport advantage (minutes per airport ride)

City drivers busy this share of paid time	City income per paid hour (USD)	Queue wait per airport ride that erases the advantage (minutes)
100% (always busy)	$68.23	10.8
75%	$51.17	27.3
60%	$40.94	43.8

Plain-language note. The breakeven wait is the queue time at which an airport driver's income per paid hour equals a city driver's. Raw data gave a smaller advantage ($13 per hour) and shorter breakeven waits (6 to 35 minutes). Data errors distorted it, such as a $4,600 fare on a 2-second trip. We report cleaned figures.

What the data cannot tell us. The O'Hare queue wait is not recorded, so the breakeven is a threshold, not a measured wait. Community area 56 is Garfield Ridge, which contains Midway but also other neighborhoods. Only 237 trips start there, so the airport group is almost entirely O'Hare. Empty driving between city fares is not recorded, so city drivers' busy share is a scenario. The board would need queue-time logs and an estimate of cruising time.

4. Which payment methods do riders use, and what should the tip screen suggest?

Answer. Most riders pay by mobile (37%) or card (36%), and 19% pay cash. Card riders tip often: 86% of card rides have a tip, with a median tip of 23% of the fare. We cannot tell whether cash riders tip, because cash tips are not recorded. A flat $3 default is wrong at both ends, at about 40% of a short fare and 5% of a long one. Offer percentage buttons, such as 20%, 25% and 30%, plus a custom amount.

Definitions. Payment type = as recorded. "Tip rate" = share of rides with a tip recorded. "Tips as % of fare" = total tips divided by total fare. Tipper = a ride with a recorded tip. Medians for tippers are shown only where at least 5 tipped rides exist.

Table 6. Payment types and tipping (cleaned data, USD unless stated)

Payment type	Share of rides	Rides with a tip	Average tip per ride (all rides)	Median tip (tippers)	Tips as % of fare
Mobile	36.6%	35.3%	$1.53	$2.94	9.5%
Credit Card	35.6%	86.1%	$6.73	$6.67	20.8%
Cash	18.9%	0.1%	$0.01	not meaningful	0.1%
Prcard	6.8%	4.6%	$0.08	$0.77	0.3%
Unknown	1.8%	0.0%	$0.00	not meaningful	0.0%
No Charge or Dispute	0.3%	0.0%	$0.00	not meaningful	0.0%

Cash: only 2 of 1,755 cash rides have a recorded tip, so the cash row describes the recording system, not rider behavior.

Figure 4. Recorded tips as % of total fare, by payment type.

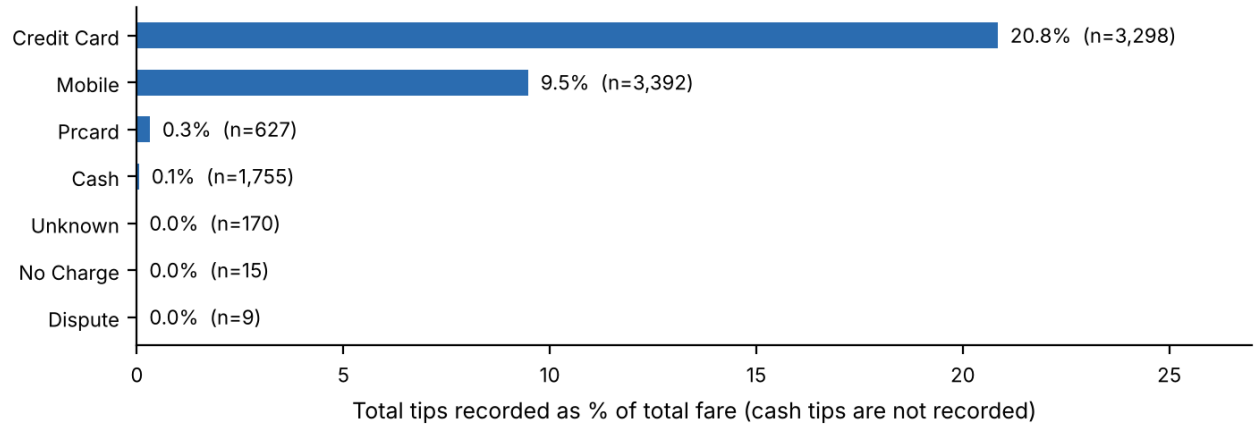


Table 7. Card tippers: a flat $3 vs a percentage default, by ride size (USD)

Card ride size (fare)	Card rides with a tip	Median fare (USD)	Median tip (USD)	Median tip as % of fare	Flat $3 as % of fare	20% of fare (median USD)
Under $10	650	$7.59	$2.00	34.0%	39.5%	$1.52
$10 to $25	599	$13.48	$3.42	24.5%	22.3%	$2.70
$25 to $50	661	$42.31	$9.55	22.5%	7.1%	$8.46
Over $50	931	$55.33	$12.16	22.0%	5.4%	$11.07

Tippers only (2,841 card rides). Median card tip across tippers: 23.1% of fare, middle half 21.0% to 29.9%.

Testing "cash riders never tip." The system recorded tips on 2 of 1,755 cash rides. Both look like entry errors ($13.15 on a $5.42 fare; $9.48 on a $15.03 fare). The data therefore cannot confirm or refute the claim. As a bound only: at the mobile rate (9.5% of fare), cash tips would total about $4,021 on $42,437 of cash fares; at the card rate (20.8%), about $8,847. These are scenarios, not measurements.

What the data cannot tell us. Whether any cash rider tips, how much, or why. Whether tippers differ by company or area. The board would need a cash tip survey or a driver-reported tip log before making any claim about cash tipping.

5. Is a fixed O'Hare-to-downtown price safe, and what should it be?

Answer. No. The average does not work, because the rule applies to each ride. At the average metered fare of $56.18, the meter beats the price on about 1 ride in 3, well above the limit of 1 in 4. Do not use $50: the meter beats it on 97% of rides. Price the trip at $59. That covers the 75th percentile of metered fare ($57.50, with a 95% confidence range up to $57.81) and every time window.

Definitions. O'Hare to downtown = pickup at O'Hare (76), dropoff in downtown (8, 32 or 33), trip of 10 or more miles. 714 rides: Near North Side 409, Loop 262, Near South Side 43. Metered fare = the fare column (excludes tips, tolls and extras). The rule: a fixed price may be below the meter on at most 1 ride in 4, so the price must be at least the 75th percentile.

Figure 5. Metered fare for O'Hare-to-downtown rides of 10+ miles (USD), with the $50 price and the 75th percentile marked.

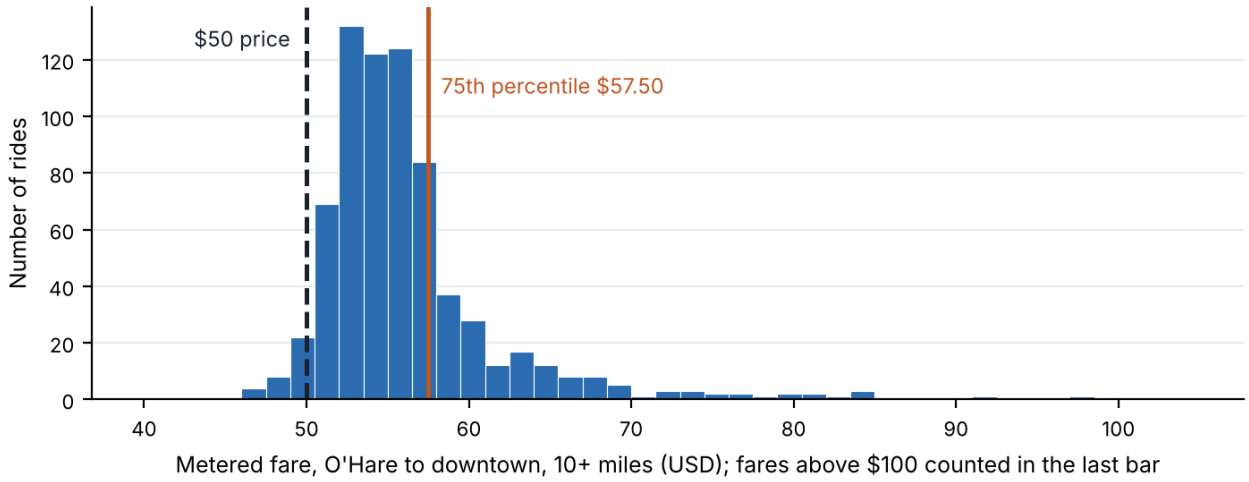


Table 8. Price test: how often the meter beats each candidate price

Candidate flat price	Share of rides where the meter is higher	Passes the 1-in-4 rule?
$50 flat price	96.8%	No
Average metered fare ($56.18)	36.0%	No
75th percentile of metered fare ($57.50)	22.5%	Yes
$59 flat price (recommended)	16.7%	Yes

"Meter is higher" means the metered fare is above the flat price, so the co-op would be paid less than the meter. The limit is 25% of rides.

Table 9. Metered fare by destination and time of day (USD)

Group	Rides	Median fare (USD)	75th percentile (USD)	Share of rides over $50
Loop	262	$55.64	$57.50	100.0%
Near North Side	409	$54.09	$56.26	94.6%
Near South Side	43	$62.77	$64.16	97.7%
Evening 16-21	223	$55.64	$58.43	96.0%
Late 21-24	99	$53.47	$55.02	96.0%
Midday 11-16	263	$55.02	$56.88	98.5%
Morning 05-11	98	$55.33	$57.96	96.9%
Overnight 00-05	31	$53.78	$54.87	90.3%

The 75th percentile is highest in the evening window ($58.43), so $59 covers every window. Near South Side has only 43 rides; its 75th percentile is $64.16. Treat that as unreliable.

Why the average is not safe. Averaging out describes total revenue, not the rule. A price at the average means the co-op is paid less than the meter on roughly half of the rides that are above average. The rule is per ride, so the test is the 75th percentile.

What the data cannot tell us. Whether riders would book at $59, and how that compares with other options. Traffic and fuel costs change over time. The sample covers one month. The board would need to decide whether Near South Side gets its own price (about $64 at the 75th percentile, but only 43 rides).

6. What can we honestly tell riders about travel time to the Loop?

Answer. Most O'Hare-to-Loop trips take 33 to 59 minutes, with a median of 48 minutes. Nine in ten trips finish within 73 minutes. Evening trips (4 to 9 pm) are the slowest. Late-night trips (9 pm onward) are the fastest of the well-sampled windows. Give riders a range, not an average.

Definitions. Trip time = the trip_seconds column divided by 60. The rounded timestamps are not used. Route = pickup O'Hare (76), dropoff Loop (32), trips of 10 or more miles. 262 rides. The typical distance is 17.8 miles.

Figure 6. O'Hare-to-Loop trip time by time of day (minutes). Each box shows the middle 50% of rides, with the median marked.

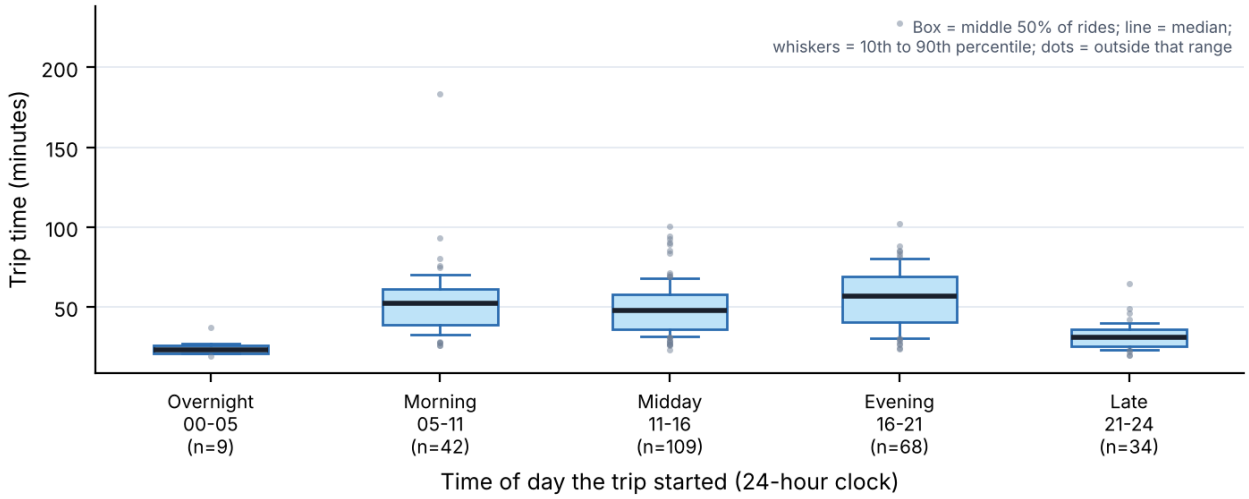


Table 10. O'Hare-to-Loop trip time by time of day (minutes)

Time of day	Rides	Median (min)	Middle 50% (min)	90th percentile (min)
Overnight 00-05	9	23.0	20.9 to 25.7	29.0
Morning 05-11	42	52.3	39.1 to 61.2	74.0
Midday 11-16	109	47.8	36.0 to 57.8	68.1
Evening 16-21	68	56.9	40.5 to 69.2	80.6
Late 21-24	34	31.1	25.2 to 36.0	41.4
All rides	262	47.7	33.1 to 59.5	72.8

Overnight (0 to 5) has only 9 rides, too few to rely on. Late night is 9 pm onward (21:00 to 23:59).

Suggested rider wording. "Most trips take 33 to 59 minutes. Allow up to 73 minutes. Evening trips can take 40 to 70 minutes, and up to about 80 at peak."

What the data cannot tell us. The wait before pickup at O'Hare is not recorded, so door-to-door time is longer. Traffic, weather and construction vary by season, and the data covers one month. The slowest trip (228 minutes) is a single outlier kept in the data. The board would need queue-time records to give riders a full door-to-door estimate.

7. What problems did we find, and what did we do about them?

Answer. We found 20 problems. We fixed 8 by removing or merging records, worked around 1 (rounded timestamps), and kept 7 with labels or without using them. We could not fix 4: the $0.50 fee gap in the total, cash tips, the sample method, and other total mismatches. Cleaning removed 731 of 10,000 trips. It did not change the answers for Sections 5 and 6, but it did change Section 3.

Definitions. "Trips affected" = the raw count for that problem. "Removed" = trips actually dropped by the cleaning rules, after earlier rules had removed some. The rules and thresholds are in the Appendix.

Table 11. Data problems, what we did, and trips affected

#	Problem found	What we did	Trips affected	Status
1	Zero-mile trips (fare cannot be checked against the meter; mostly cancelled)	Dropped all zero-mile trips	624	Fixed (624 removed)
2	Fare below $3.25 (meter start)	Dropped (rule R1)	69	Fixed (6 more removed after R0)
3	Fare above 3 times the meter and above $50 (includes a $4,600 cash entry)	Dropped (rule R2)	67	Fixed (8 more removed)
4	Fare below 40% of meter on trips over 2 miles	Dropped (rule R3)	9	Fixed (6 more removed)
5	Trips under 60 seconds	Dropped (rule R4)	450	Fixed (87 more removed)
6	Trips over 4 hours	Rule R5; all 3 already caught by R3	3	Fixed (0 extra)
7	Speed over 80 mph	Rule R6; all 3 already caught by R3 or R4	3	Fixed (0 extra)
8	Company name with double space	Merged into one name	79	Fixed (merged)
9	Truncated company names ("Medallion Leasin")	Left as is; company is not used in any answer	244	Kept, not used
10	Taxi IDs recorded under more than one company	Left as is; flagged for Section 7 note; not used	101 taxis	Kept, not used
11	Two taxis with overlapping trips	Kept; no rule covers overlaps	2 pairs	Kept
12	Timestamps rounded to 15 minutes (start = end on short trips)	Used trip_seconds for all durations	2,269	Worked around; limit remains
13	trip_total is $0.50 above the components on card and mobile rides	Not used for income, per board decision	5,948	Could not fix
14	Other trip_total mismatches ($1 to $3)	Not used for income	83	Could not fix
15	Cash tips not recorded by the system	Stated as a limit in Sections 1, 4 and 7	1,755 cash rides	Could not fix
16	Unexplained payment types (Unknown, Dispute, No Charge, Prepaid)	Kept and labeled; no tips recorded on these	226	Kept, labeled; limit remains
17	Blank pickup community area	Labeled "Outside Chicago" (an assumption)	258	Labeled; limit remains
18	Blank dropoff community area	Left blank; affects only the airport and city split	829	Kept
19	Rides with no fare or tips	Kept; excluded from income sums	3	Kept
20	Sample drawn by an unknown method	Assumed representative, per board direction	All	Could not fix

Table 12. Key results before and after cleaning (USD unless stated)

Key result	Raw data	Cleaned data	Changed?
Income per ride, typical (median)	$16.12	$16.61	Small
Income per ride, average (mean)	$27.59	$27.29	Small
Airport income per hour (USD)	$88.06	$87.29	Small
City income per hour (USD)	$75.38	$68.23	**About $7 lower**
Airport advantage per hour (USD)	$12.68	$19.07	**Larger**
Breakeven queue wait, city always busy (min)	6.3	10.8	**Larger**
O'Hare-to-downtown 75th percentile fare (Section 5)	$57.50	$57.50	None
O'Hare-to-Loop median trip time (Section 6)	47.7 min	47.7 min	None
Card tips as % of fare (Section 4)	21.3%	20.8%	Small

Raw figures use the same definitions on the unfiltered file. The airport advantage and breakeven waits are the only material changes, and the direction of the Section 3 conclusion holds either way.

Appendix

A. Cleaning rules, applied in this order

Rule	Threshold	Why
R0. Zero miles	miles = 0	Fare cannot be checked against the meter; mostly cancelled trips
R1. Meter start	fare below $3.25	Lower than the meter can charge at the start
R2. Extreme fare	fare above 3 times meter estimate and above $50	Catches entry errors, such as $4,600
R3. Low fare on long trips	trips over 2 miles with fare below 40% of meter estimate	Fare too low to match distance
R4. Very short trip	trip under 60 seconds	Likely cancelled or erroneous
R5. Very long trip	trip over 4 hours	Likely an error
R6. Impossible speed	speed over 80 mph	Likely an error

Meter estimate = $3.25 + $2.25 per mile + $0.20 per 36 seconds of trip time. The 106-mile, $301 trip is kept: its fare matches the meter, and it lasted under 4 hours.

B. Method notes

- **Income:** fare + tips. Trip time: trip_seconds (not the rounded timestamps). Miles: trip_miles.
- **Hourly income (Section 3):** total income divided by total trip hours, for each group. Breakeven wait: solved so that airport income per paid hour equals city income per paid hour, where city paid hours = meter hours divided by busy share.
- **Confidence interval (Section 5):** bootstrap. We resampled the 714 O'Hare-to-downtown rides with replacement 10,000 times, computed the 75th percentile each time, and took the middle 95% of those values. Random seed fixed at 42, so the result repeats.
- **Percentiles:** computed on individual rides, without weighting.
- **Time windows:** based on the start hour of each trip.

C. Files

- Cleaned data: taxi_trips_sep2026_clean.csv (9,269 trips). Cleaning script: clean_trips.py. Cleaning report: data_cleaning_report.md.
- All figures in this report are generated from the cleaned data by make_figs.py and build_report.py in the same working folder.